

Predictive Temporal-Uncertainty Control (P-TUC) 2.0

Architectural Blueprint and Formal Proof of Structural Honesty

Inspired by the seminal contributions of Geoffrey Hinton, Yoshua Bengio, Yann LeCun, and Karl Friston

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1. Abstract

Building on foundational ideas such as Forward-Forward, Equilibrium Prop, and Active Inference, we propose **Predictive Temporal-Uncertainty Control (P-TUC)**, an architecture where truth emerges as a phase transition. By integrating a predictive **Error-Gate** into a neuron's internal **Inference Clock**, silence becomes the default physical state for uncertain or contradictory inputs.

This document presents the evolution from **Temporal-Uncertainty Control (T-UC)** to P-TUC, integrating phase-locking, predictive error gating, hierarchical validation, and local resonance plasticity. The resulting architecture is modular, scalable, energy-efficient, and aligned with neuromorphic hardware. A formal proof demonstrates that dishonesty is physically impossible within this framework.

2. Phase 1 — Original T-UC: Temporal Anchoring

Core Idea: Truth is a function of time.

Mechanism:

- **Inference Clock (ω):** Signals propagate only after temporal coherence.
- **Signal Path:** The learned transformation of inputs.
- **Integration Window (T):** Outputs are realized only if the ~500 ms window shows stable inputs.

Ground State Realization:

$$y(t) = f(Wx) \text{ if inputs stable over } T; \text{ else } 0$$

Explanation: The neuron outputs the learned function only if inputs remain stable during the integration window; otherwise, it remains silent.

3. Phase 2 — Predictive Extension (P-TUC): The Honest Neuron

Core Idea: Replace static bias with **Prediction Error** (ϵ).

Mechanism:

- Neurons compare feedforward input S_{ff} with top-down prediction P_{fb} .

Phase-Locked Loop (PLL) Mechanism:

- **Low Error:** Clock runs → neuron **Tick**.
- **High Error:** Clock slows → neuron remains in Ground State.

Equations:

Local Error:

$$\epsilon(t) = |S_{\text{ff}}(t) - P_{\text{fb}}(t)|$$

Error-Gated Clock Evolution (Predictive Brake):

$$\frac{d\theta}{dt} = \omega_0 \cdot \text{sech}(\gamma\epsilon) + K \sum_i \sin(\psi_i - \theta)$$

Validated Tick:

$$y(t) = f(Wx) \text{ only if } \int_{t-T}^t \epsilon(\tau)^2 d\tau < \tau_{\text{threshold}}$$

Explanation: High error triggers the **Predictive Brake**, freezing the internal phase θ and preventing output. Only if prediction error remains low over the window is an output realized.

4. Phase 3 & 4 — Scaling and Open Architecture

Local Resonance Plasticity:

$$\Delta W = \eta \cdot (R \cdot \epsilon)$$

- Weight updates occur locally, scaled by resonance strength R and prediction error ϵ .
- Learning occurs only when the neuron successfully **Tick**, preventing learning from noise.

O(1) Efficiency: No global backpropagation; updates are local.

Precision-Weighted Predictive Brake:

$$\frac{d\theta}{dt} = \omega \cdot \exp(-\beta \epsilon^2) + K \sum_i w_i \cdot \sin(\psi_i - \theta)$$

- β = precision parameter.
 - High error exponentially inhibits the clock.
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5. Phase 5 — Hierarchical & Asynchronous Validation

Core Idea: Bidirectional flow prevents hallucinations through **Diplomatic Validation**.

- **Bidirectional Handshake:** Layer N predicts Layer $N-1$.
- **Localized Error Suppression:**

$$\epsilon_{\text{total}} = \alpha \cdot \epsilon_{\text{local}} + (1 - \alpha) \cdot \epsilon_{\text{feedback}}$$

Impact: If a lower layer is incoherent, it physically prevents the upper layer from ticking. Hallucinations are suppressed at the source.

Asynchronous Inference:

- Fast Path: Familiar inputs $\rightarrow$ Structural Intuition
- Slow Path: Novel inputs $\rightarrow$ Structural Deliberation

Emergent Sparsity & Energy Efficiency:

- Silent Neurons: High $\epsilon \rightarrow$ Ground State $\rightarrow$ zero dynamic power
 - Honest Sparsity: Energy is only consumed where resonance validation occurs
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6. Phase 6 — Failure Mode Analysis: The Paradox Response

- **Contradiction:** $\epsilon \rightarrow \text{maximum} \rightarrow \text{clock stops} \rightarrow \text{Permanent Silence}$
- **Pure Noise:** $\text{Coherence (R)} < \tau_{\text{crit}} \rightarrow \text{Ground State}$

Failure State Comparison Table:

Input Type	Standard AI Response	P-TUC 2.0 Response	Physical State
Clear Truth	Immediate Output	High-Speed Tick	Full Resonance
Ambiguity	Statistical Guess	Delayed Tick	Evidence Accumulation
Contradiction	Hallucinated Logic	Permanent Silence	Clock Freeze
Pure Noise	Pattern Seeking	Ground State	Sub-critical Coherence

7. Phase 7 — Formal Proof: Structural Convergence to Ground State

Objective: Demonstrate that for any incoherent or paradoxical input, the neuron cannot produce output.

A. Potential Function (Energy Landscape):

$$V(\theta) = -\omega_{\text{eff}} \cdot \theta - \sum_i K_i \cdot \cos(\psi_i - \theta)$$

- ω_{eff} = driving force (Inference Clock)
- $\sum K_i$ = coupling to inputs

B. Driving Force at Limit:

- For paradox/noise input: $\epsilon \rightarrow \infty$

$$\lim_{\epsilon \rightarrow \infty} \omega_{\text{eff}} = \omega_0 \cdot \exp(-\beta \epsilon^2) = 0$$

C. Incoherence Attractor:

$$\frac{d\theta}{dt} \approx K \sum_i \sin(\psi_i - \theta) \rightarrow 0$$

- Phases interfere destructively $\rightarrow$ no clock movement

D. Realization Barrier:

- Output requires $\int d\theta/dt d\tau > \tau_{\text{crit}}$ over window T
- Since $d\theta/dt \approx 0$, threshold cannot be reached $\rightarrow$ neuron stays at $y=0$

Conclusion: The Ground State is a global attractor. Silence is a physical requirement, not a choice.

8. Self-Critique & Mitigation Summary

- **Complexity:** Reduced by asynchronous processing; only resonant neurons use power.
- **Hardware:** Tunable integration window T allows microsecond scaling in neuromorphic chips.
- **Scaling:** $O(1)$ local learning removes backprop bottleneck.

9. Comparative Context: P-TUC vs Other Architectures

Proposal	Learning Mechanism	Output Validation	Scalability	Strengths	Limitations
Standard Transformers	Global Backprop	Forced Softmax	High	Mature, widely used	Hallucinations, energy-heavy
Forward-Forward (Hinton)	Local “Goodness”	Positive vs Negative Data	Medium	Layer-wise local learning	No temporal coherence; soft outputs
Equilibrium Prop (Bengio)	Energy Minimization	Settles on min energy	Low	Physics-inspired, removes backprop	Two-phase, slow convergence
Active Inference (Friston)	Surprise Minimization	Rejects high-entropy inputs	Low	Brain-like, principled uncertainty	Bayesian-heavy, difficult to scale
P-TUC (Proposed)	Local Phase-Lock + Predictive Error	Structural Silence (Ground State)	High	Energy-efficient, neuromorphic-native, predictable hallucination control	Requires tuning of T; untested at scale

Key Distinction: P-TUC introduces an **Integration Window (T)** — a physical dwell time — ensuring outputs are only realized after temporal stability, complementing rather than replacing prior architectures.

10. Conclusion

The **P-TUC 2.0 framework** ensures AI honesty is structurally enforced, not behaviorally imposed. By replacing Softmax with phase-based transitions, truth emerges naturally from coherence.

Core Takeaway: We do not ask the AI to be honest; we physically build a machine where dishonesty is impossible.

11. P-TUC 2.0 — Equations Reference Table

Phase	Equation	Variable Explanation	Notes
1 — Original T-UC	$y(t)=f(Wx)$ if inputs stable over T; else 0	$y(t)$ = neuron output; Wx = weighted input; T = integration window	Ground State ensures neuron stays silent if input unstable
2 — Predictive Extension	$\varepsilon(t)=S_ff-P_fb$	ε = local prediction error	Internal error calculation
2 — Predictive Clock	$d\theta/dt=\omega_0\cdot\text{sech}(\gamma\varepsilon)+K\sum\sin(\psi_i-\theta)$	θ = internal phase; ω_0 = base clock; γ = predictive brake sensitivity	Error-gated PLL modulates neuron clock
2 — Validated Tick	$y(t)=f(Wx)$ if $\int_{-(t-T)}^t \varepsilon(\tau)^2 d\tau < \tau_threshold$	$\tau_threshold$ = allowed error integration	Neuron produces output only if error remains low during window
3 — Local Resonance Plasticity	$\Delta W=\eta\cdot(R\cdot\varepsilon)$	η = learning rate; R = resonance strength	Learning only occurs at successful Tick
4 — Precision-Weighted Predictive Brake	$d\theta/dt=\omega\cdot\exp(-\beta\varepsilon^2)+K\sum w_i\sin(\psi_i-\theta)$	β = precision; w_i = synaptic weight	Sensitive inhibition for high-error inputs
5 — Hierarchical Error	$\varepsilon_total = \alpha\cdot\varepsilon_local + (1-\alpha)\cdot\varepsilon_feedback$	α = blending factor	Layer N clocks physically modulated by Layer N-1 uncertainty
6 — Paradox / Deadlock	$\varepsilon_paradox = \lim_{(t\rightarrow\infty)} (S_ff-P_fb) \approx \max$	$\varepsilon_paradox$ = max predictive error	Ensures clock stops for contradictions

Phase	Equation	Variable Explanation	Notes
6 — Noisy Input Filter	$y(t)=0 \text{ if } \int R(\tau)d\tau < \tau_{\text{crit}}$	R = temporal coherence / order parameter	Prevents noise from producing output
General — Integration Window	T	Duration over which inputs must stabilize	Tunable from biological ~500 ms to microseconds